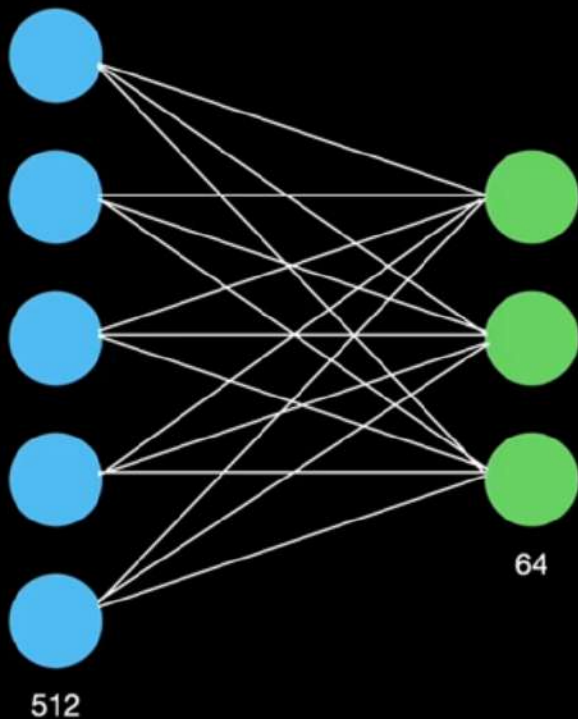


$$(E_{\text{Apple}} \cdot W_k)$$

Linear Transformation

$$\begin{matrix} (64, 1) & (64, 512) & (512, 1) \\ \swarrow & \uparrow & \searrow \\ A = W \cdot X \end{matrix}$$

1. Changing Dimensional
2. Extracting richer features

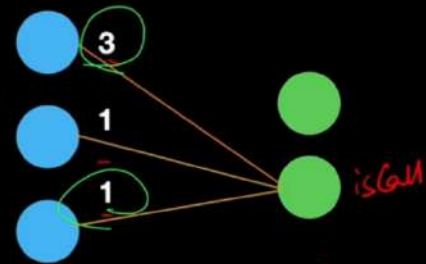
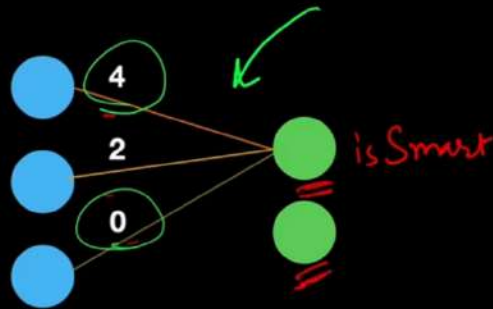


$$\begin{bmatrix} 0.12 \\ 0.23 \\ \vdots \\ 0.23 \end{bmatrix}$$

$$(64, 1)$$



	Apple	Orange	Watch
isTech	2	0	1
isWatch	0	0	3
isFruit	2	3	0



$$\begin{bmatrix} 2 & 0 & 2 \end{bmatrix} \times \begin{bmatrix} 4 & 3 \\ 2 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 8 \end{bmatrix}$$

$$2 \times 4 + 0 \times 2 + 2 \times 0$$

$$2 \times 3 + 0 \times 1 + 2 \times 1$$

Apple

$$\begin{bmatrix} 0 & 0 & 3 \end{bmatrix} \times \begin{bmatrix} 4 & 3 \\ 2 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 \end{bmatrix}$$

Orange

$$\begin{bmatrix} 1 & 3 & 0 \end{bmatrix} \times \begin{bmatrix} 4 & 3 \\ 2 & 1 \\ 0 & 1 \end{bmatrix}$$

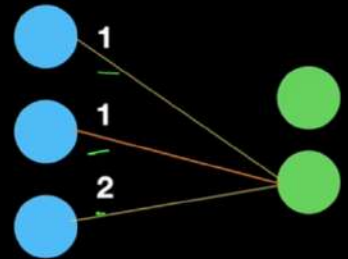
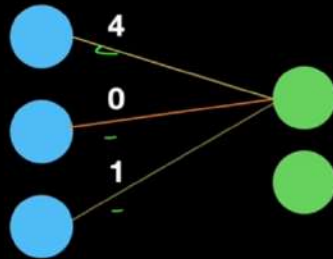
$$\begin{bmatrix} 10 & 6 \end{bmatrix}$$

Watch

$$\underline{Q_{watch}} = [10, 6]$$

	Apple	Watch
isTech	2	1
isWatch	0	3
isFruit	2	0

W_k



$$\underline{K_{watch}} = \underline{E_{watch}} \times \underline{W_k}$$

$$\underline{K_{apple}} = \underline{E_{apple}} \times \underline{W_k} \rightarrow 64$$

$$(\underline{Q_{watch}} \times \underline{K_{watch}}) \leftarrow$$

$$(\underline{Q_{watch}} \times \underline{K_{apple}}) \leftarrow 136$$

$$\underline{K_{apple}} = [2, 0, 2] \times \begin{bmatrix} 4 & 1 \\ 0 & 1 \\ 1 & 2 \end{bmatrix} = [10, 6]$$

$$\underline{K_{watch}} =$$

$$[1, 3, 0] \times \begin{bmatrix} 4 & 1 \\ 0 & 1 \\ 1 & 2 \end{bmatrix}$$

$$= [4, 4]$$

